

Causal Discovery in Cancer Gene Expression Data using Bayesian Networks

NAME: PRAAVIN KUMAR MANICKAM SRINIVASAN

MAIL-ID: pmanicka@uci.edu

1. Introduction

Breast cancer begins when DNA mutations cause normal cells to divide uncontrollably independent of estrogen. In ER Positive patients estrogen connects to the Estrogen Receptor (ER) protein on cancer cells and accelerates their growth. Hormone therapy blocks this receptor and works effectively. In ER Negative patients cancer cells produce little or no ER protein leaving hormone therapy with nothing to block and completely failing. The ESR1 gene produces the ER protein. Standard machine learning finds genes that correlate with ER Status but cannot identify causation. This project applies Bayesian Networks to RNA-seq gene expression data from 549 breast cancer patients from TCGA to identify which gene causally determines ER protein production and therefore ER Status.

2. Data and Preprocessing

The TCGA Breast Cancer dataset contained 705 patients with 1941 columns. After removing patients with unclear ER Status 549 remained: 414 ER Positive and 135 ER Negative. RNA-seq expression columns (rs_prefix, 604 genes) were used as input with ER Status as the binary outcome (Positive=1, Negative=0). The natural class imbalance of 414 vs 135 patients reflects real-world ER Status distribution. Synthetic data generation (SMOTE) was deliberately avoided as creating artificial cancer patient data would introduce bias into the causal analysis.

Gene Filtering: 604 → 80 Genes

- Filter 1 — MSigDB Curated List: 604 genes cross-referenced against the Molecular Signatures Database of known breast cancer genes. Result: 371 biologically validated genes.
- Filter 2 — Limma Empirical Bayes: Linear model fitted for each gene: $\text{Expression} = \beta_0 + \beta_1 \times \text{ER_Status} + \epsilon$. Moderated t-statistic: $t = (\text{mean ER}^+ - \text{mean ER}^-) / \text{moderated SE}$, where SE is stabilised using information from all 371 genes simultaneously. Benjamini-Hochberg FDR correction applied: BH threshold = $(\text{Rank}/371) \times 0.05$. Result: 314 genuinely significant genes.
- Top 80 genes with smallest adjusted p-values selected balancing biological coverage with computational feasibility.

Discretization

Gene values converted to Low / Medium / High categories using 33rd and 66th percentile cutoffs applied independently per gene. Three categories chosen to avoid boundary problems of two categories while maintaining approximately 183 patients per category for reliable BDe scoring. Final dataset: 549×81 (80 genes + ER Status).

3. Methods

Hill Climbing with BIC

Starts with an empty network. Adds, removes or reverses one arrow at a time keeping only changes improving $\text{BIC} = \log P(\text{Data}|\text{Network}) - (k/2) \times \log(N)$. The probability term measures how consistently each arrow holds across 549 patients. The penalty term $(k/2) \times \log(N)$ costs 3.15 BIC units per parameter preventing overfitting. Limitation: can get stuck at local maxima.

Tabu Search with BDe

Same as Hill Climbing but maintains a Tabu List of forbidden recent moves forcing exploration of new directions and escaping local maxima. Always returns the best solution found. $\text{BDe} = \prod_i \prod_j [\Gamma(\alpha_{ij}) / \Gamma(\alpha_{ij} + N_{ij})] \times \prod_k \Gamma(\alpha_{ijk} + N_{ijk}) / \Gamma(\alpha_{ijk})$ adds Dirichlet prior counts $\alpha_{ijk} = N' / (q_i \times r_i)$ preventing overconfident probability estimates from small sample counts unlike BIC.

Bootstrap Validation: 500 Replicates

549 patients resampled with replacement 500 times. New network built each time. Two metrics reported: Strength = connections appearing in >50% of 500 networks (reliable). Direction = proportion of networks agreeing on arrow direction. Higher values closer to 1.0 indicate more confident causal direction while values near 0.5 indicate uncertain direction. Bootstrap run separately for both algorithms.

Consensus Network

Final network built keeping only arrows appearing in >50% of bootstrap networks. Hill Climbing: 131 → 86 arrows. Tabu Search: 114 → 82 arrows.

4. Results

Bootstrap Strength and Direction

- ESR1 → ER Status (Hill Climbing): Strength = 0.876, Direction = 0.953. Highly reliable and directionally confident.
- ESR1 → ER Status (Tabu Search): Strength = 0.894, Direction = 0.928. Both algorithms independently confirm ESR1 → ER Status making this an extremely robust finding.
- AGTR1 → ER Status rejected: Strength = 0.342 below 50% threshold. Too unstable to be considered reliable.
- FOXA1 ↔ ESR1: Strength = 0.952 (very strong) but Direction = 0.418 vs 0.582 (58/42 split). Reported as undirected edge as direction cannot be determined from observational data alone.

Consensus Network Visualizations

Both consensus networks show ESR1 (orange) as the only direct parent of ER Status (red). FOXA1 (green) strongly connected to ESR1 with uncertain direction.

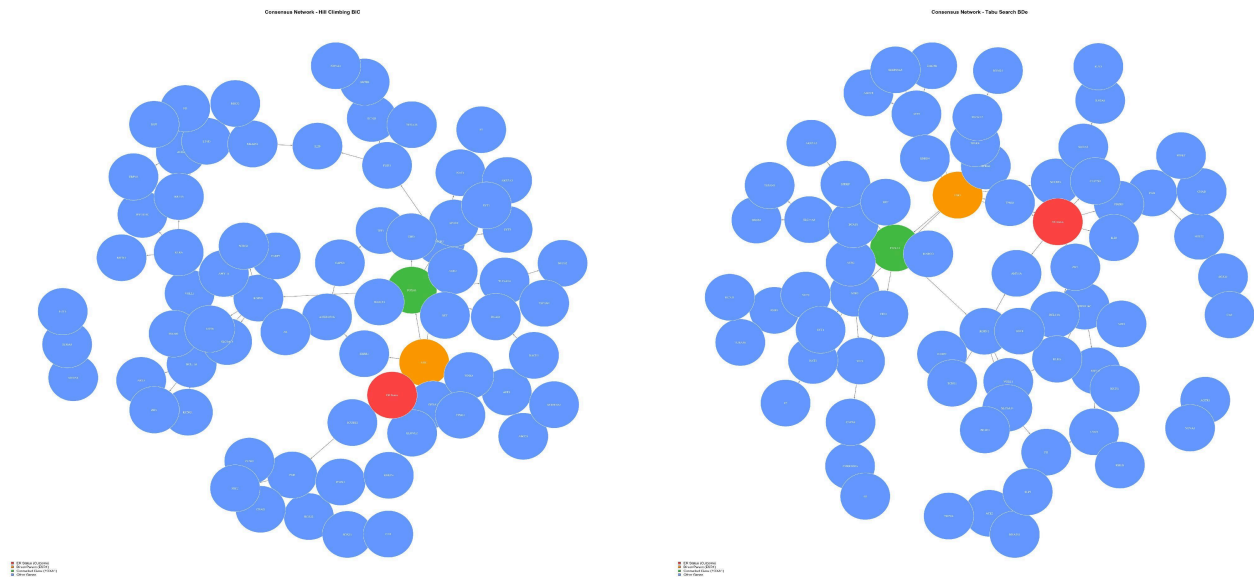


Figure 1: Hill Climbing Consensus Network (left) and Tabu Search Consensus Network (right)

Validation: BBN vs LASSO Logistic Regression

5-fold cross validation using same 80 genes and 549 patients for all three models:

Model	Fold 1	Fold 2	Fold 3	Fold 4	Fold 5	Average
Hill Climbing BBN	91.8%	86.4%	87.3%	90.0%	87.2%	88.5%
Tabu Search BBN	91.8%	86.4%	90.0%	90.0%	86.2%	88.9%
LASSO	99.1%	93.6%	92.7%	85.5%	89.0%	92.0%

BBN performs only 3-4% below LASSO but provides causal insight LASSO cannot. LASSO predicts who will be ER Negative. BBN predicts who will be ER Negative AND identifies ESR1 as the causal gene providing a potential therapeutic target.

5. Biological Validation

The ESR1 finding was validated against published literature confirming it as a known driver of breast cancer treatment resistance:

- Bidard et al. 2025 (New England Journal of Medicine): ESR1 mutations are the most common mechanism of acquired resistance to aromatase inhibitor therapy in advanced breast cancer directly confirming ESR1 as a critical determinant of hormone therapy response. DOI: 10.1056/NEJMoa2502929
- Razavi et al. 2018 (Cancer Cell): Genomic sequencing of 1918 breast cancers found ESR1 genomic alterations enriched in endocrine resistant tumors and FOXA1 alterations also identified as enriched in endocrine resistant tumors directly validating both our ESR1 finding and the FOXA1 connection. DOI: 10.1016/j.ccell.2018.08.008

Both independent publications in top-tier journals confirm that ESR1 is a genuine and clinically significant determinant of treatment resistance in breast cancer validating our causal discovery finding.

6. Limitations

- Observational data cannot definitively prove causation. All findings are causal hypotheses for experimental follow-up. Causal sufficiency and faithfulness assumptions may not fully hold as TCGA does not capture every confounder.
- 549 patients instead of proposed 1100 affects direction confidence particularly for upstream genes.
- FOXA1 and ESR1 show 95.2% bootstrap strength but direction remains uncertain at 58/42 split and is reported as undirected.
- Expression data alone cannot capture ESR1 mutations which are a known driver of acquired treatment resistance that emerges during hormone therapy. Our analysis measured ESR1 expression activity not DNA mutations. Future work should integrate mutation data (mu_ columns) alongside expression data for a more complete analysis.

7. Future Work

- Incorporate mutation profiles (mu_ columns) with RNA-seq metrics to better identify treatment resistance drivers that gene expression levels fail to capture.
- Expand the sample size toward the original 1100-patient target to enhance bootstrap directionality confidence for genes upstream of ESR1.
- Conduct laboratory validation such as knockdown assays to confirm the ESR1 → ER Status link suggested by this observational study.
- Analyze longitudinal expression patterns across multiple treatment stages to uncover the biological mechanisms regulating ESR1 variability among breast cancer patients.
- Broaden the Bayesian framework to a multi-omics approach integrating copy number and mutation data for a holistic view of cancer resistance.

References

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2. Razavi P et al. The Genomic Landscape of Endocrine-Resistant Advanced Breast Cancers. Cancer Cell. 2018;34(3):427-438. DOI: 10.1016/j.ccell.2018.08.008